TSM 098

Homework 6

Linear equations, parentheses, and fractions

Topics 24-26, 28 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

Work without a calculator before checking. Give exact answers. For equations with fractions, combine and simplify each side first. Then move each remaining denominator to multiply the entire opposite side. Check solutions in the original equation.

Isolate the variable

1. Solve and check.

$-7x = 21$

2. Solve and check.

$\frac{2}{3}x = -8$

3. Solve and check.

$5x - 4 = 2x + 11$

4. Solve and check.

$7 - 3x = 2x - 8$

Parentheses and solution sets

5. Solve. State a value, no solution, or all real numbers as appropriate.

$3(x - 2) + 4 = 2x + 9$

6. Solve. State a value, no solution, or all real numbers as appropriate.

$5 - 2(3x - 4) = x + 6$

7. Solve. State a value, no solution, or all real numbers as appropriate.

$4(x + 2) - 3 = 4x + 5$

8. Solve. State a value, no solution, or all real numbers as appropriate.

$2(x - 1) + 3 = 2x + 5$

9. Solve. State a value, no solution, or all real numbers as appropriate.

$3(2x - 1) - 2(x + 4) = x + 7$

Fraction equations

10. Combine and simplify each side first, then solve and check.

$\frac{x}{3} + \frac{1}{2} = \frac{5}{6}$

11. Combine and simplify each side first, then solve and check.

$\frac{x+2}{4} = \frac{2x-1}{3}$

12. Combine and simplify each side first, then solve and check.

$\frac{3x-1}{5} - \frac{x}{2} = \frac{7}{10}$

13. Combine and simplify each side first, then solve and check.

$\frac{2x+5}{6} + \frac{x}{4} = \frac{3}{2}$

14. Combine and simplify each side first, then solve and check.

$\frac{x+1}{3} - \frac{x-2}{6} = 2$

Explain a correction

15. A student turns 2(x - 3) = 10 into 2x - 3 = 10. Explain and correct the distribution error, solve, and check.